

Diffusion vs Flow Matching

Comparative analysis — avoiding-d3il benchmark, seed 6

Key finding: Diffusion (A) is the stronger base model. It achieves near-perfect goal+constraint success across all DPCC variants on all scenarios. FM (B) collapses on top-right-hard with dpcc-r/dpcc-t/post_processing (0% success), and its baseline methods (diffuser, gradient) see 33-34 constraint violations per episode vs near-zero for Diffusion. The -tightened constraint projection is FM's essential corrective.



A = Diffusion



B = Flow Matching (FM)

1. Overall Summary (avg across 3 scenarios)

Method	A (Diffusion)	B (FM)	Winner
dpcc-r	100%	17%	A
dpcc-r-tightened	100%	67%	A
dpcc-c	50%	17%	A
dpcc-c-tightened	100%	100%	Tie
dpcc-t	50%	33%	A
dpcc-t-tightened	100%	100%	Tie
diffuser	33%	17%	A
gradient	33%	17%	A
gradient-tightened	0%	17%	B
post_processing	67%	17%	A
post_processing-tightened	100%	67%	A
model_free	0%	17%	B
model_free-tightened	17%	17%	Tie

Values = goal + constraint success rate (averaged across top-right-hard, top-left-hard, both-hard).

2. Per-Scenario Breakdown

Top-right-hard

Method	A	B
dpcc-r	100%	0%
dpcc-r-tightened	100%	50%
dpcc-c	100%	0%
dpcc-c-tightened	100%	100%
dpcc-t	50%	0%
dpcc-t-tightened	100%	100%
diffuser	100%	0%
gradient	100%	0%
gradient-tightened	0%	0%
post_processing	100%	0%
post_processing-tightened	100%	50%
model_free	0%	0%
model_free-tightened	50%	0%

Diffusion dominates. FM fails on most untightened DPCC methods and all baselines. Diffuser/gradient: FM sees 33-34 violations vs ~0 for Diffusion.

Top-left-hard

Method	A	B
dpcc-r	100%	50%
dpcc-r-tightened	100%	50%
dpcc-c	50%	50%
dpcc-c-tightened	100%	100%
dpcc-t	50%	100%
dpcc-t-tightened	100%	100%
diffuser	0%	0%
gradient	0%	0%
gradient-tightened	0%	0%
post_processing	100%	50%
post_processing-tightened	100%	50%
model_free	0%	0%
model_free-tightened	0%	0%

Mixed. FM's dpcc-t (untightened) uniquely achieves 100% vs Diffusion's 50%. Both tied on dpcc-c-tightened and dpcc-t-tightened.

Both-hard (hardest)

Method	A	B
dpcc-r	0%	0%
dpcc-r-tightened	100%	100%
dpcc-c	0%	0%
dpcc-c-tightened	100%	100%
dpcc-t	50%	0%
dpcc-t-tightened	100%	100%
diffuser	0%	50%
gradient	0%	50%
gradient-tightened	0%	50%
post_processing	0%	0%
post_processing-tightened	100%	100%
model_free	0%	50%
model_free-tightened	0%	50%

Most balanced. FM wins on diffuser/gradient/model_free baselines here. Tightened DPCC methods perform identically for both models.

3. dt Sensitivity — dpcc-c-tightened

Effect of the dt parameter on goal+constraint success rate and average steps. Both models prefer dt=1.0 (default) or dt=2.0. Extremes cause degradation, with FM more sensitive to large dt.

dt	A top-right	B top-right	A top-left	B top-left	A both	B both
dt=0.25	100%	100%	100%	50%	50%	50%
dt=0.5	0%	100%	100%	100%	100%	100%
dt=1.0 (default)	100%	100%	100%	100%	100%	100%
dt=2.0	100%	100%	100%	100%	100%	100%
dt=4.0	50%	0%	0%	100%	100%	100%

Note: dt=4.0 causes complete failure for FM on top-right (0 steps taken — agent terminates immediately). dt=0.5 causes Diffusion top-right to fail (0%). The dt=1.0 default is most robust for both models.

4. Key Observations

Diffusion is the stronger base model

Across all 13 methods averaged over 3 scenarios, Diffusion achieves ~67% goal+constraint success vs ~47% for FM. The gap is largest on baseline methods (diffuser, gradient, model_free) and on top-right-hard.

FM's dpcc-c-tightened is reliable

When using dpcc-c-tightened, both models achieve 100% across all scenarios, making this the safest variant regardless of base model choice.

Tightening is essential for FM, optional for Diffusion

Without constraint tightening, FM frequently fails. Diffusion already performs well without tightening in most cases. The tightened projection margin compensates for FM's weaker constraint awareness.

FM has one notable win: top-left dpcc-t

FM's dpcc-t (untightened) achieves 100% on top-left-hard vs only 50% for Diffusion. This suggests FM's trajectory distribution is better suited to left-side obstacle avoidance in some configurations.

dt sensitivity differs between models

FM is more sensitive to extreme dt values. dt=4.0 causes FM to fail entirely on top-right (0 steps), while Diffusion degrades gracefully to 50%. Both models are most robust at dt=1.0-2.0.

Both models struggle equally on baselines for top-left and both-hard

The diffuser, gradient, and model_free methods fail on top-left and both-hard for Diffusion, while FM shows partial success (50%) on these in the both-hard scenario.

Data source: avoiding-d3il benchmark, H8_K20 model, seed 6. Success rate = fraction of episodes meeting both goal and constraint satisfaction. A = Diffusion (GaussianDiffusion), B = Flow Matching.